

Project Analytics: Benefits, Challenges and First Steps

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Underperformance is nothing new in the world of capital projects. Large projects experience cost and schedule overruns consistently. They are not meeting their intended objectives, not delivering expected business value. We've been hearing the same story for years.

Additionally, projects are beginning to have a greater impact on shareholder value. This has raised the stakes and made it imperative to find a way to manage projects more effectively.

So the question is, what are we going to do about it? The answer may be buried in our data. What if we could leverage data to make the best possible decisions? And not just reactive decisions. What if we could identify and fix potential issues before they turn into project-crushing overruns? That is the promise of project analytics.

In the following article, we will explore project analytics, their benefits, where many organizations are failing, and some practical steps to better leverage project analytics to improve project outcomes.

What are Project Analytics?

Project analytics are practices of systematically analyzing data to obtain information that helps you make better decisions. By applying statistical models to your data, you can gain key insights that you wouldn't be able to otherwise.

In an environment where projects are becoming increasingly complex, project analytics are a lifeline for project managers to stay on schedule and on budget. Using analytics, project managers can move beyond simply capturing data. They can see exactly how projects are performing, and whether or not they are delivering intended business benefits. Project analytics can also be predictive, giving you insight into what is likely to happen on a project, and inform the best actions to take.

In summary project analytics are a [valuable tool for project managers](#) to make strategic decisions and improve project success rates.

A Data Problem = A Project Analytics Opportunity

More than [90% of all data](#) has been created in the last two years – with around 40 trillion gigabytes of data in existence. That is mind boggling.

This trend holds true when it comes to capital projects. Technology has enabled more complex projects to be executed by more people from more places. And with digitalization, more and more data is captured throughout the project lifecycle.

With the boom of data creation, most businesses have come to a logical conclusion. There has to be a way to leverage all of this data to improve performance. In fact, 83% of Engineering & Construction executives feel that organizations will be [data-driven within the next 5 years](#), including routine use of data analytics and predictive modeling for project planning.

The leaders in the industry clearly see project analytics as an opportunity, but are they capitalizing?

Where Are We Now? The Status Quo

More data is captured about a project than ever before, but there is limited access to real-time insights, so the value has not been unlocked.

The inability to quickly and efficiently report on project progress means many companies are only doing backward looking reporting. For generations, project teams have utilized data to [produce reports that inform what has happened.](#)

When I first joined the project controls world, my mentor told me “It’s not your job to tell them the license plate of the bus that hit them, but to tell them that the bus is coming.”

– Garrett Fultz, Senior Managing Director, FTI Consulting

The project is analyzed at its completion in an attempt to determine how to improve the next time around. However, there often isn’t a clear picture of why a project performed the way it did, or what can be done about it.

Overall, the utilization of data has not been good enough. Consistently high project performance can only be achieved by better leveraging project analytics, helping to avoid issues before they arise.

The Benefits of Project Analytics

When employed properly, analytics can have a big impact on project outcomes. I recently interviewed [Garrett Fultz](#), Senior Managing Director at [FTI Consulting](#) about the promise of project analytics and how organizations can leverage data to improve decision making and increase project predictability.

Project analytics can help deliver value in a variety of areas, including:

Benchmarking

Project analytics help you evaluate project performance against [both internal and external datasets](#). Meaning you can evaluate your organization relative to internal factors like project team, department, business unit etc. or to measure your company relative to others. As you capture more information about projects, you can analyze more dimensions to find areas for improvement.

Forecasting Accuracy and Predictability

Anybody can tell you at the end of the project how it performed in relation to the original budget and schedule. But the real value is [how early in the project can you accurately forecast the outcome](#). Project analytics can identify issues earlier in the project, giving you an opportunity to take corrective action. By avoiding late surprises, you will be able to avoid overruns.

Time-series Analysis

Project analytics also help you identify trends over particular time periods. You can see if you are improving over time relative to a specific set of variables. This can be helpful in determining if new standards or process adjustments are having intended impacts.

Ad Hoc Analyses

Often, decision makers are faced with a specific challenge or question that needs to be answered. Project analytics support decision makers with answers to specific questions when they need it – providing the evidence needed to make decisions more confidently.

5 Levels of Project Analytics Maturity

Project analytics exist to help you answer questions. The more mature project analytics practices become, more complex questions can be answered. Lets take a look at the five levels of [project analytics maturity](#):

1. Descriptive Analytics

The first level of maturity is descriptive analytics. They can do backward-looking reporting, answering basic questions like what happened? Or when did it happen? Most of the reports that business generate fall in this category. It is the status quo discussed earlier.

2. Diagnostic Analytics

The next level is being able to analyze past performance and answer why did it happen? If you can effectively gather project data, analytics can help you interpret, identify anomalies, detect patterns and determine relationships between things like cost and performance.

3. Predictive Analytics

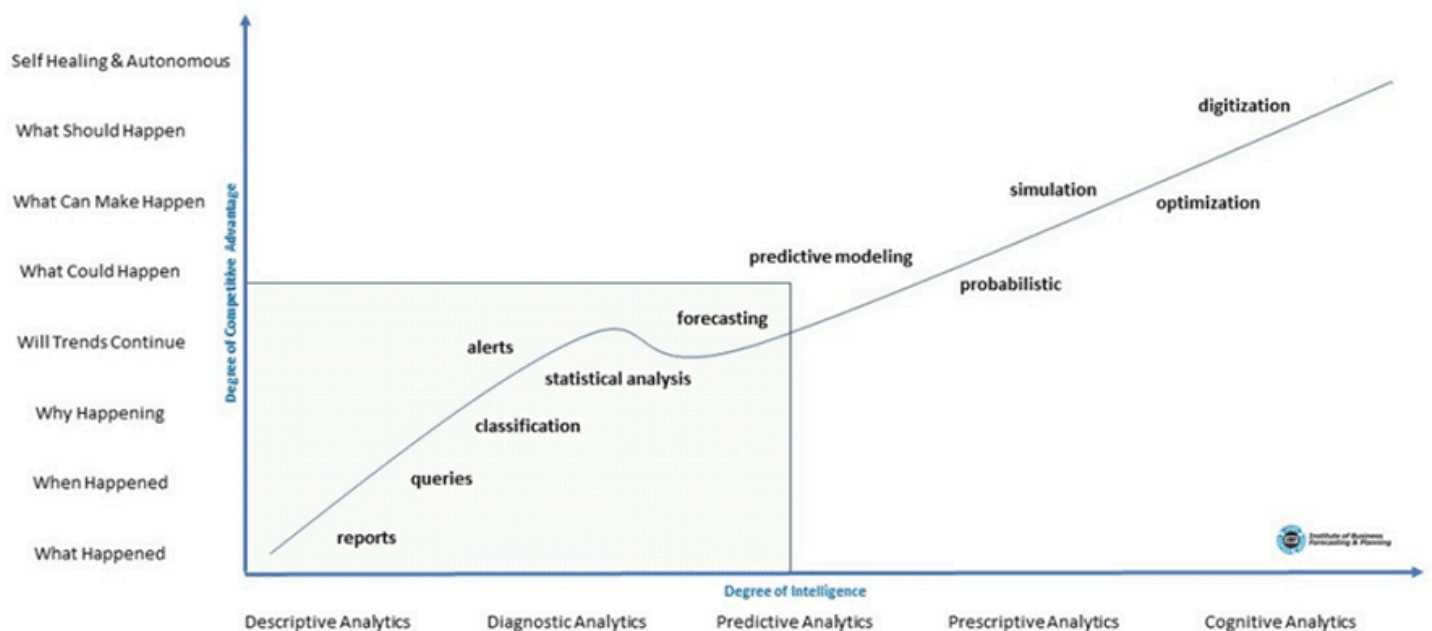
This level is where you begin to get forward looking, answering questions like what will likely happen? Applying statistical analysis and predictive models can use past performance to determine likely outcomes for the future. These early warning signs help decision makers prevent small issue from becoming large ones.

4. Prescriptive Analytics

The next level takes the information provided by predictive analytics and uses it to answer the question what should we do? This is continuous automated improvement, meaning as you implement corrective actions and measure the impacts of those actions, the statistical model can learn from those outcomes to suggest the best path forward.

5. Cognitive Analytics

As you apply the previous levels of analytics, you can begin to get very sophisticated and answer questions like what don't we know? This involves using machine learning and artificial intelligence (AI) to have your statistical models define new models.



Challenges Leveraging Project Analytics

With all the data that is captured during a project, the assumption is sometimes that it will be easy to utilize – that predicting issues, making better decisions and improving

outcomes will happen by default. Instead, projects are doing worse. Why is that?

Siloed and Inaccurate Data

One reason is the sources of project data are siloed. By using so many different point solutions, spreadsheet and in-house tools to capture project data, project teams are becoming more inefficient. As a result, they often spend most of their time chasing down data, compiling it, and correcting errors. Without a centralized source of accurate project data, it will be hard for organizations to move beyond the *Descriptive* level of maturity.

Unstructured Data

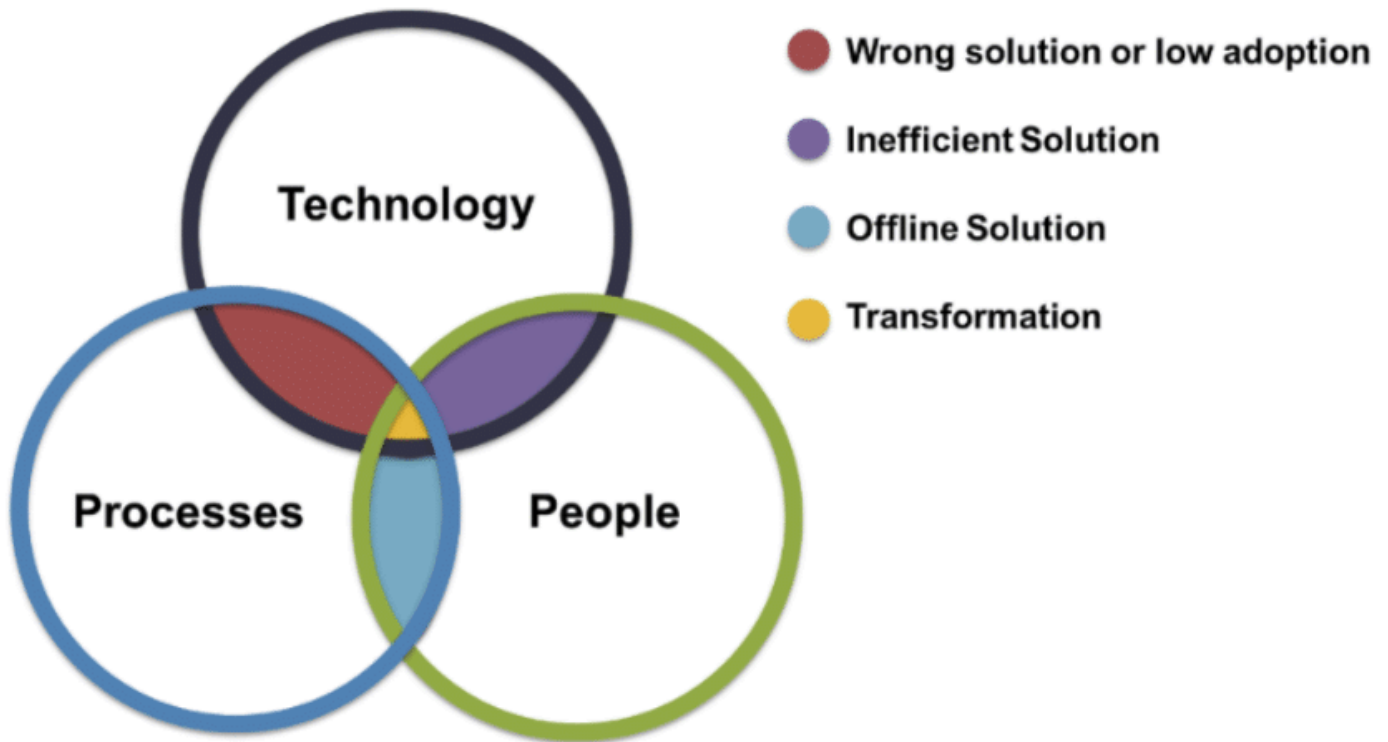
Another problem much of the data is unstructured, meaning it exists in different forms and is not clearly defined. This is a big problem for businesses because it makes data difficult and costly to manage, organize and utilize for decision making. In fact, [95% of businesses cite unstructured data as a major challenge](#). This can lead businesses to pursue AI and machine learning solutions to make sense of their data, attempting to jump straight to the highest levels of maturity.

Lack of a Good Project Controls and Project Management Foundation

However, this creates another potential problem. If organizations focus solely on the technology to analyze data while neglecting to utilize best practice processes for project management and project controls, efforts around AI and machine learning will fall short of expectation.

It would be like building a super speed train but forgetting to build the track.

Success lies at the intersection of people, processes, and technology. Project analytics should be designed based on the capability of the right technology platform as the source of project data. A platform based on industry good practice will ensure project teams are capturing and leveraging data in a standardized way.



Only by having immediate access to accurate and up-to-date project information, AND a good foundation of project management and [project controls](#) processes, will project teams be able to fully leverage analytics to deliver the best possible outcomes.

Practical Steps to Improve Project Performance with Project Analytics

So where to start? Here are some tips on your journey to unlock the power of your data.

1. Define Your Vision

As with any transformation, the first step is defining a vision for where you want to go. The vision should be comprehensive and reimagine how your data can unlock more value.

2. Adopt a Foundational Platform

Another important step is to adopt a centralized project performance management solution. This will help you consolidate your project data, break down siloes of

information, and build the necessary processes to leverage your data. Having connectivity between project data and achieving a single source of truth can help you answer descriptive and diagnostic maturity level questions efficiently and accurately. It will also provide the basis for you to build more mature project analytics.

3. Start Small

Any transformation will be doomed to fail if it is too forceful so it is important not to go too big right away. Your vision should be rolled out in iterations. Start by answering one question that you couldn't answer before, or more easily than you could answer before. Once you begin to better understand your data, you can begin to better understand your business. It's important to see project analytics as a journey – one that's going to add incremental value every time you deploy a new model or a new analytic.

4. Achieve Quick Time to Value

People expect results quickly. As you move through your journey, it is important to answer questions that will deliver immediate impacts to the business. With these new insights, you may be able to determine other areas of impact, or other data that needs to be collected.

5. Make User Adoption Easy

It doesn't matter if you have the most powerful tool in the world if nobody will use it. You need willing adoption. Companies that make digital tools accessible, self-service, and part of standard operating procedures are [much more likely to succeed](#). If your project analytics depend on experts to find insights, you are limiting its potential. A strong, intuitive data visualization platform will allow decision makers to get what they need more quickly.

6. Build on Your Successes

As you show results, you will get more buy-in and support. Then, you will have the justification for more investment. As you build momentum and confidence in your project analytics you can begin to get more mature, answering more questions. At that point, you

can layer an integrated analytics platform on top of your project data source to provide fast, scalable data models.

How Enterprise Project Performance Software Can Help

Enterprise Project Performance (EPP) software combines [project portfolio management](#), [project controls](#) and [project management](#) software all in one platform. This makes it the ideal foundation to help build your organization into a data analytics powerhouse.

Enterprise project performance software replaces a multitude of commercial point solutions, homegrown tools, Excel spreadsheets, Access databases, and SharePoint sites – giving you a centralized source for all project related data like:

- Project Metadata
- Project Personnel
- Contractor Information
- Cost Data
- Schedule Data
- Commitments & Changes
- Variances

Breaking down siloes of information allows project teams to break free from the status quo – spending more time leveraging project analytics, rather than chasing down data and correcting errors.

The right EPP platform will help you achieve level 1 and 2 project analytics maturity and start to measure data around level 3. As a result, you will be able to quickly answer ‘*what happened?*’ and ‘*why did it happen?*’ and starting understand ‘*what will likely happen?*’

Then, you can add analytical models on top the existing data in your EPP. This will help you move to level 3 and beyond.

Implementing project analytics on top of [the right EPP platform](#), built around good project management processes will result in better insights for decision makers and ultimately better project performance.